

Interim Information Design

Hershdeep Chopra ^{*} Jeffrey C. Ely [†]

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Abstract

We study efficient and consumer-surplus maximizing information policies in a bilateral trade setting where the buyer is initially privately imperfectly informed about his willingness to pay. We show that providing more information to the buyer can lead to higher market prices and a lower trade probability without affecting the consumer or producer surplus.

Introduction

We revisit the information design problem in a single-item bilateral trade environment where the buyer (he) and the seller (she) are initially uninformed and have a common prior for the buyer's valuation, as studied in [Roesler and Szentes \(2017\)](#) (henceforth RS). The seller is a monopolist and dictates the terms of trade by posting a price. A designer provides information to the buyer about his valuation. The seller knows the signal structure of the information provided by the designer, but only the buyer observes the signal realization. The designer shapes the buyer's valuation of the item through information and thus changes the demand faced by the seller.

We analyze information design in a similar bilateral trade model from an *interim* perspective. Before the designer can release information to the buyer, the buyer observes an (exogenous) noisy signal about his valuation for the item. The buyer's interim valuation constitutes his payoff-relevant private information, and we will

^{*}Department of Economics, Northwestern University. hershdeepchopra2026@u.northwestern.edu.

[†]Department of Economics, Northwestern University. jeff@jeffely.com.

refer to it as the buyer's *type*. In particular, we study information design when the buyer and seller have heterogeneous priors and the designer can provide information conditional on the buyer's type. The distribution of buyer types is the *interim demand* faced by the seller. We ask how market outcomes, like probability of trade, prices, and welfare, are affected by information provision when the buyer has access to an additional exogenous signal. In particular, we study how the designer's information policy can shape an *interim demand* into a *market demand* which determines the price.

RS characterizes all possible equilibrium consumer-producer surplus pairs in their model. They show that only efficient outcomes occur in equilibrium, and efficient outcomes entail trade with probability 1. As a result, in RS, the market price completely determines the consumer-producer surplus pair for any efficient outcome.

We characterize all possible *efficient* consumer-producer surplus pairs when the buyer is partially informed about his valuation. An *efficient* outcome generates the maximum possible surplus from trade. In particular, an outcome is efficient if and only if a buyer does not buy the item only if his value for the item is lower than the seller's value for the item (which we normalize to zero). We show that the designer can provide information to turn any inefficient *interim demand* into a *market demand* that strictly improves surplus for both the buyer and the seller. In this case, the designer can provide information to the buyer and improve trade efficiency.

In our model for any given efficient consumer-producer surplus pair, there could be many different corresponding market prices. This extra flexibility between welfare and prices depends on the designer's ability to generate information conditional on the buyer's type.

Our main contribution is a welfare-independent comparative static result between prices and buyer information. Fixing some efficient consumer-producer surplus pair, the corresponding prices increase as the buyer is better informed about his valuation. This highlights an important friction involved with information provision in such environments; a more informed buyer makes a better purchase decision, but also faces higher prices. Importantly, we show that more informed buyers can drive up prices without affecting the welfare.

A result central to our analysis is the identification of a class of iso-elastic demands that are sufficient to generate all possible efficient consumer-producer surplus pairs. Moreover, any efficient equilibrium demand can be transformed into a demand in this family through some information policy, chosen by the designer,

without affecting the welfare and price.

Finally, we provide a necessary condition for an outcome to be buyer optimal and characterize the buyer optimal outcome with the highest price.

Model

We study a bilateral trade model with a buyer (he) and a monopolist seller (she). The seller offers a single item, produced at no cost. We normalize the seller's reservation value to 0, hence trade is efficient. The underlying value of the product is binary and represented by $\theta \in \{0, 1\}$. At the onset, the buyer privately observes a (exogenous) signal about θ , represented by his prior belief $\mu \in [0, 1]$. We refer to the buyer's private belief μ as his type. The buyer's type is his expected valuation for the item (or his willingness to pay).

There is a commonly known distribution $F \in \Delta([0, 1])$ from which types are drawn.¹ The distribution F is the interim demand that the seller faces. The a designer announces an information policy, denoted by $\rho(\cdot) = (\rho_0(\cdot), \rho_1(\cdot))$, where for each μ

$$\rho_\theta(\mu) \in \Delta([0, 1])$$

Here $\rho(\mu)$ represents a statistical experiment about θ whose realization is only observed by type μ buyer. More precisely, if the item's value is θ , then the type μ buyer observes the realization of $\rho_\theta(\mu)$. In response to the announced information policy, the seller forms a posterior distribution (*posterior demand*) G about the distribution of types and offers a price p to the buyer. After observing the outcome of the information policy, the buyer decides to purchase the item at a price p or not.

The offered price is profit-maximizing

$$p \in \operatorname{argmax}_{s \in [0, 1]} s(1 - G(s) + \Delta(G, s))$$

Where $\Delta(G, s)$ represents the size of the atom at s for the distribution G .

$$\Delta(G, s) := \lim_{x \downarrow s} G(x) - \lim_{x \uparrow s} G(x)$$

¹This amounts to a commonly known distribution for the buyer's exogenous signal and its correlation with θ .

The designer can induce a distribution G using some information policy $\rho(\cdot)$ if and only if $G \succeq_{mps} F$. Where $G \succeq_{mps} F$ holds whenever for every real valued continuous convex function φ we have (see [Elton and Hill \(1992\)](#) theorem 4.7 for details)

$$\int_0^1 \varphi(s) dG(s) \geq \int_0^1 \varphi(s) dF(s)$$

By Theorem 3.A.1 in [Shaked and Shanthikumar \(2007\)](#), the above is equivalent to the following

$$\int_0^p G(s) ds \geq \int_0^p F(s) ds \text{ for all } p \in [0, 1]$$

And with equality at $p = 1$.

We will refer to such a distribution G as a *feasible distribution* (or *feasible demand*). Similar to RS, we will work with *posterior demand* G instead of the underlying information policy.

Feasible Demands

As pointed out in the previous section, given a prior F , the set of feasible seller's beliefs about the distribution of buyer's valuations (type) is given by

$$\mathcal{G}(F) := \{G : [0, 1] \rightarrow [0, 1] \mid G \succeq_{mps} F, G \text{ is a cdf} \}$$

In the interim setting, the buyer is partially informed, and the designer can provide information about the θ type-by-type. To each buyer type μ , there corresponds a distribution of a posterior expected valuation resulting from the information provided by the designer. This posterior distribution is the market demand of type μ buyer. The market demand faced by the seller is the average across types μ , with respect to F , of these type contingent market demands.

Whereas in RS the buyer and seller have a common prior F over the buyer's valuation. The designer announces an information policy which maps a buyer valuation $v \in [0, 1]$ to a distribution over signals whose realization is only observed by the buyer. Unlike the interim setting, in RS, the buyer has no additional information, so no targeting is involved in information provision. Thus, the market demand is given by the distribution of the expected posterior valuation of the buyer with prior F . This set of distributions is the set of feasible market demands and is given by

$$\mathcal{G}_{UI}(F) := \{G : [0, 1] \rightarrow [0, 1] \mid F \succeq_{mps} G, G \text{ is a cdf} \}$$

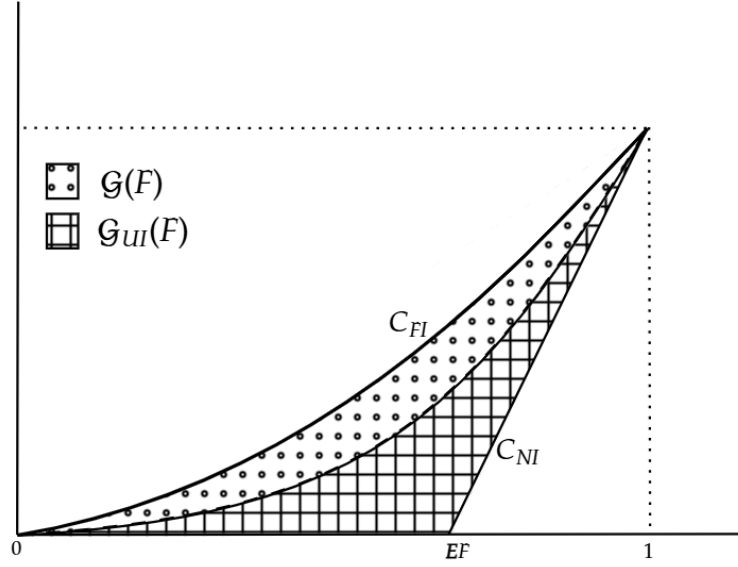


Figure 1: Feasible Demands

We can visualize this difference concisely using the information functional representation from [Gentzkow and Kamenica \(2016\)](#) (see [Figure 1](#)). Define the following convex functions

$$C_F(\mu) = \int_0^\mu F(v)dv, \quad C_{NI}(\mu) = \int_0^\mu 1_{\{v \geq EF\}} dv, \quad C_{FI}(\mu) = \mu(1 - EF)$$

The above are the information functionals defined for prior F , Dirac mass on EF , and a fully informed buyer, respectively. The region between C_{NI} and C_{FI} constitutes all possible information functionals generated by some demand with expected valuation EF . The functional C_F partitions this region into demands that are mean preserving spread of F , given by $\mathcal{G}(F)$, and demands that are mean preserving contraction of F , given by $\mathcal{G}_{UI}(F)$.

Equilibrium Outcomes

An *equilibrium profile* comprises of the tuple (G, p) such that 1) $G \in \mathcal{G}(F)$, 2) $p \in \operatorname{argmax}_{s \in [0,1]} s(1 - G(s) + \Delta(G, s))$.

Each equilibrium profile has its corresponding *outcome* (κ, π) where κ is the con-

sumer surplus and π is the producer surplus given by

$$\kappa := \int_p^1 (s - p) dG(s)$$

$$\pi := p(1 - G(p) + \Delta(G, p))$$

In particular, $\pi \leq p$ and $\kappa \leq \mathbf{EF}$.

Efficient Demands

An equilibrium profile is *efficient* if it generates the maximum total surplus, i.e. $\kappa + \pi = \mathbf{EF}$. As trade always generates a positive surplus, efficiency requires inducing the maximum probability of trade while accounting for the buyer's additional information. The following lemma characterizes efficient demands that can be generated by information design.

Lemma 1. *An equilibrium profile (G, p) is efficient if and only if $\text{supp}(G) \cap (0, p) = \emptyset$.*

Proof. Fix some equilibrium outcome (G, p) . If $\text{supp}(G) \cap (0, p) = \emptyset$ then conditional on $\theta = 1$ the trade happens with probability one. The sufficiency part of the claim follows from the law of total expectation.

For necessity, consider $\text{supp}(G) \cap (0, p) \neq \emptyset$, then we can construct a new cdf. The designer can provide additional information to all buyers with willingness to pay in $(0, p)$. For any type $\mu \in (0, p)$, the additional information induces a posterior valuation 0 with probability $1 - \alpha$, a valuation p with probability $(1 - \delta)\alpha$, and valuation $p + \varepsilon$ with probability $\delta\alpha$. The mean preserving constraint then requires $\alpha = \frac{\mu}{p + \delta\varepsilon}$.

The proof is complete if there exists $\varepsilon, \delta > 0$ such that the seller still finds it optimal to set the price p . We have increased the mass of buyers with valuation p by $\alpha(1 - \delta) \int_0^p dG(s)$. The mass of buyers willing to buy at a price $p_1 \in (p, p + \varepsilon]$ is increased by $\alpha\delta \int_0^p dG(s)$. The producer surplus from charging a price above $p + \varepsilon$ is unchanged. As G was an outcome, we only need to verify whether the transformed demand creates incentives to price at some $p_1 \in (p, p + \varepsilon]$. Under the transformed demand, the seller's payoff from charging price p is

$$\pi + p\alpha(1 - \delta) \int_0^p dG(s)$$

The seller's payoff from charging a price $p_1 \in (p, p + \varepsilon]$ is bounded above by

$$\pi + (p + \varepsilon)\alpha\delta \int_0^p dG(s)$$

The profit maximizing price is p whenever $p \frac{1-2\delta}{\delta} \geq \varepsilon > 0$. $\square$

Note that the construction in the proof of [Lemma 1](#) strictly increases both the consumer and producer surplus, thus an **equilibrium profile is efficient if and only if it is on the Pareto frontier**. Thus, it's without loss to focus on efficient outcomes if one is interested in determining producer or consumer optimal equilibrium outcomes. In general, many different demands G can be feasible for the same price p and lead to the same equilibrium outcome. These demands could be incomparable with respect to the convex order. We will focus on the following efficient demands determined by a market price p , a producer surplus π , and an expected valuation m .

$$H_p^\pi(s; m) = \begin{cases} 1 - \frac{\pi}{p} & s < p \\ 1 - \frac{\pi}{t(p, \pi, m)} & s = p \\ 1 - \frac{\pi}{s} & 1 > s \geq t(p, \pi, m) \\ 1 & s = 1 \end{cases}$$

Where $t(p, \pi, m)$ is implicitly defined as the solution to the following whenever it exists ²

$$\pi + \int_t^1 \frac{\pi}{s} ds + p\pi \left[\frac{1}{p} - \frac{1}{t} \right] = m \quad (1)$$

Whenever $m = \text{EF}$ we will drop m from the notation and just write H_p^π and $t(p, \pi)$. Note for later that the consumer surplus κ from any implementable H_p^π is given by $\kappa = \text{EF} - \pi$.

Define

$$\mathcal{H}(F) := \{G \in \mathcal{G}(F) \mid \exists \pi, p \text{ such that } G = H_p^\pi\}$$

These demands are iso-elastic (or unit elastic) on their support. For a given $H \in \mathcal{H}(F)$, the seller is indifferent between charging any positive price on the support of H . We will show that the above set of demands is sufficient to generate all possible efficient outcomes. Moreover, any efficient equilibrium demand can be

²If $p < 2 - \frac{m}{\pi}$, then [Equation 1](#) holds for no $t < 1$. In this case if $\pi \leq m < \pi(2 - p)$ we let $t(p, \pi) := 1$ and the distribution H_p^π is described by three mass points $H_p^\pi(0) = 1 - \frac{\pi}{p}$, $\Delta(H_p^\pi, 1) = \frac{m - \pi}{1 - p}$, and $\Delta(H_p^\pi, p) = 1 - \Delta(H_p^\pi, 1) - H_p^\pi(0)$.

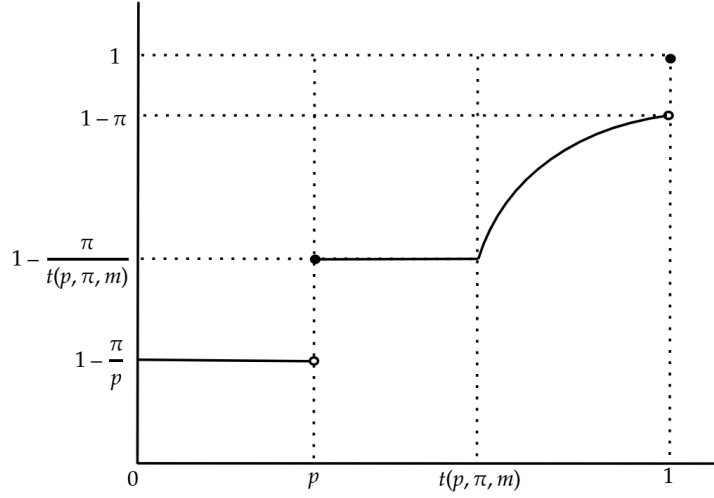


Figure 2: $H_p^\pi(s; m)$ when Equation 1 holds for $t \leq 1$

transformed into a demand from $\mathcal{H}(F)$ through some information policy without affecting the welfare and price.

Lemma 2. *If (G, p) is an efficient equilibrium profile with corresponding outcomes (κ, π) then (H_p^π, p) is an outcome equivalent equilibrium profile. Moreover $H_p^\pi \succeq_{mps} G$.*

Proof. First we show the claim for a profile (G, p) with producer surplus π such that $p \leq 2 - \frac{EF}{\pi}$. Let $\tilde{G}$ be a CDF such that $\Delta(\tilde{G}, 1) = \frac{EF - p \int_p^1 dG(s)}{1-p}$, $\Delta(\tilde{G}, p) = \frac{\int_p^1 dG(s) - EF}{1-p}$, and $\tilde{G}(0) = G(0)$. The CDF $\tilde{G}$ is constructed by type-by-type garbling all types μ in the interval $[p, 1]$ into posteriors supported on $\{p, 1\}$, thus $\tilde{G} \succeq_{mps} G$. As $p \leq 2 - \frac{EF}{\pi}$ it must be that $\Delta(\tilde{G}, 1) \leq \pi$. and $p = \operatorname{argmax}_{[0,1]} s(1 - \tilde{G}(s) + \Delta(\tilde{G}, s))$. In particular, $(\tilde{G}, p)$ is an equilibrium profile with producer surplus π . Note that as (G, p) is an equilibrium profile it must be that $\pi \leq EF$, thus $\tilde{G} = H_p^\pi$.

Let $p > 2 - \frac{EF}{\pi}$. As (G, p) is an equilibrium profile, we get that $p \in \operatorname{argmax}_s s(1 - G(s) + \Delta(G, s))$. This implies that $G(s) \geq 1 - \frac{\pi}{s}$ for all $s \in [p, 1]$ and $\Delta(G, 1) \leq \pi$. Thus the curve G lies weakly above the curve $H_p^\pi(\cdot; \pi(1 - \ln p))$. Where $\pi(1 - \ln p)$ is the maximum possible expected valuation of a demand for which the profit maximizing price is p and which delivers a producer surplus of π . As (G, p) is an

outcome we get that $\mathbf{EG} = \mathbf{EF} \leq \pi(1 - \ln p)$. By assumption $(2 - p)\pi < \mathbf{EF}$. Thus there exists a unique $t(p, \pi) \in [p, 1]$ which satisfies Equation 1 and H_p^π is well defined.

As both G and H_p^π are efficient demands for producer surplus π and price p we get that $G(0) = H_p^\pi(0) = 1 - \frac{\pi}{p}$. Since $G(s) \geq 1 - \frac{\pi}{s}$ for all $s \in [p, 1]$, we get $\frac{\pi}{p} - \frac{\pi}{t(p, \pi)} \geq \Delta(G, p)$ as otherwise G lies strictly above H_p^π on $(p, 1)$ which can not happen as G and H_p^π have the same mean. Moreover, as G lies weakly above H_p^π on the interval $[t(p, \pi), 1]$, Equation 1 (mean is preserves) along with $G(0) = H_p^\pi(0)$ further imply that there exists $p_1 \in (p, t(p, \pi))$ such that $G(s) < H_p^\pi(s)$ for $s \in [p, p_1)$ and $G(s) \geq H_p^\pi(s)$ for $s \in [p_1, t(p, \pi))$. In particular, we have shown G crosses H_p^π once and from below.

The above, along with Equation 1 implies that $H_p^\pi \succeq_{mps} G$. As (G, p) is an equilibrium profile we get that $H_p^\pi \succeq_{mps} F$. By construction $p \in \arg\max_{s \in [0, 1]} s(1 - H_p^\pi(s) + \Delta(H_p^\pi, s))$, thus (H_p^π, p) is an equilibrium profile. It induces the same producer surplus π , price p , probability of trade $1 - \Delta(G, 0)$ and consumer surplus $\mathbf{EF} - \pi$. $\square$

When buyer and seller have a common prior, the efficient outcome involves trade with probability one at a price below the prior expected valuation. Thus, the outcome is determined by the equilibrium price. An efficient outcome isn't completely determined by the equilibrium price when the buyer is partially informed, as the probability of trade might be bounded away from 1. Moreover, multiple prices can support the same efficient outcome.

Fixing an outcome, how does the change in the amount of information affect the equilibrium prices? The result below highlights comparative statics between the amount of information provided to the buyer and the price level in equilibrium, keeping the expected consumer and producer surplus fixed at some efficient level. This might be relevant, say, for a policymaker who is concerned about some exogenous spill-over effect of high prices on the general economy, as well as concerned about unnecessarily intervening.

Theorem 1. *For any $p_0 < p_1$ and π , if $(H_{p_0}^\pi, p_0)$ and $(H_{p_1}^\pi, p_1)$ are equilibrium profiles then $H_{p_1}^\pi \succeq_{mps} H_{p_0}^\pi$.*

Proof. By definition to prove $H_{p_1}^\pi \succeq_{mps} H_{p_0}^\pi$ it suffices to show that $t(p_0, \pi) \geq$

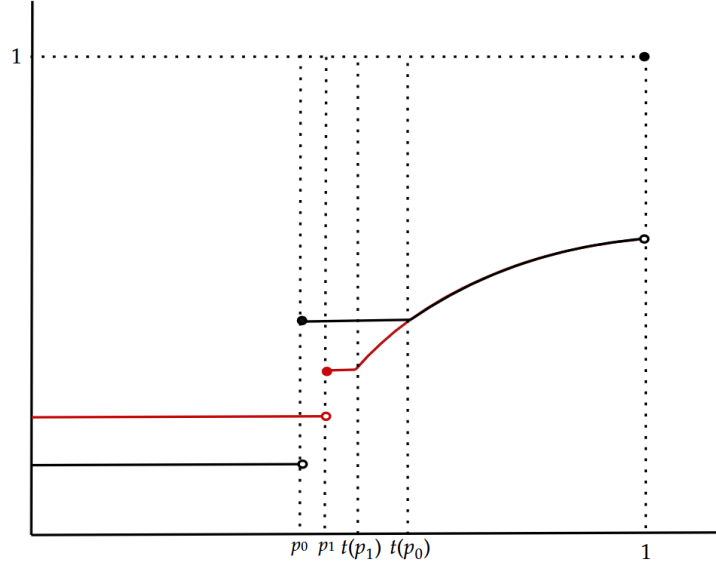


Figure 3: Single crossing between $H_{p_0}^\pi$ in black and $H_{p_1}^\pi$ in red.

$t(p_1, \pi)$. By Equation 1 we get that

$$\begin{aligned}
 & p_0 \left(\frac{\pi}{p_0} - \frac{\pi}{t(p_0, \pi)} \right) + \int_{t(p_0, \pi)}^1 \frac{\pi}{s} ds = p_1 \left(\frac{\pi}{p_1} - \frac{\pi}{t(p_1, \pi)} \right) + \int_{t(p_1, \pi)}^1 \frac{\pi}{s} ds \\
 \implies & p_0 \left(\frac{\pi}{p_0} - \frac{\pi}{t(p_0, \pi)} - \frac{\pi}{p_1} + \frac{\pi}{t(p_1, \pi)} \right) = (p_1 - p_0) \left(\frac{\pi}{p_1} - \frac{\pi}{t(p_1, \pi)} \right) + \int_{t(p_1, \pi)}^{t(p_0, \pi)} \frac{\pi}{s} ds \\
 \implies & p_0 \left(\frac{\pi}{p_0} - \frac{\pi}{p_1} \right) + p_0 \int_{t(p_1, \pi)}^{t(p_0, \pi)} \frac{\pi}{s^2} ds = (p_1 - p_0) \left(\frac{\pi}{p_1} - \frac{\pi}{t(p_1, \pi)} \right) + \int_{t(p_1, \pi)}^{t(p_0, \pi)} \frac{\pi}{s} ds \\
 \implies & p_0 \left(\frac{\pi}{p_0} - \frac{\pi}{p_1} \right) = (p_1 - p_0) \left(\frac{\pi}{p_1} - \frac{\pi}{t(p_1, \pi)} \right) + \int_{t(p_1, \pi)}^{t(p_0, \pi)} \left(\frac{\pi}{s} - p_0 \frac{\pi}{s^2} \right) ds \\
 \implies & \frac{p_1 - p_0}{t(p_1, \pi)} = \int_{t(p_1, \pi)}^{t(p_0, \pi)} \left(\frac{1}{s} - p_0 \frac{1}{s^2} \right) ds
 \end{aligned}$$

As $t(p_1, \pi), t(p_0, \pi) \geq p_0 \geq 0$ we get that the above equation holds only if $t(p_0, \pi) \geq t(p_1, \pi)$. $\square$

More informed buyers are less (more) likely to trade, conditional on the item having value $\theta = 0$ ($\theta = 1$). Thus, keeping the total surplus and the share of surplus fixed, the seller sets a higher price p for more informed buyers.

Buyer Optimal

In this section, we determine the buyer optimal outcome. From [Lemma 1](#), we know that the buyer optimal outcome must be efficient. The following lemma presents a sufficient condition under which the consumer surplus can be strictly improved.

Proposition 1. *Given some efficient equilibrium profile (G, p) with outcome (κ, π) , if $p < 2 - \frac{EF}{\pi}$ then there exist $p_0 < p$ and $\pi_0 < \pi$ such that $H_{p_0}^{\pi_0} \succeq_{mps} G$ and $(H_{p_0}^{\pi_0}, p_0)$ is an equilibrium profile.*

Proof. By the argument in [Lemma 2](#), we can restrict attention to the distribution H_p^π . Define the following functions $\Pi(s) := s(1 - H_p^\pi(0))$, and $W(s) := \Delta(H_p^\pi, 1) + \frac{p-s}{1-s} \Delta(H_p^\pi, p)$. As $p < 2 - \frac{EF}{\pi}$ and (H_p^π, p) is an equilibrium profile we get that $W(p) < \Pi(p)$ and $W(0) > \Pi(0)$. Thus, by continuity and the intermediate value theorem, there is a price $0 < p_0 < p$ such that $\Pi(p_0) = W(p_0)$.

Now define a CDF $\tilde{G}$ such that $\tilde{G}(0) = H_p^\pi(0)$, $\Delta(\tilde{G}, p_0) := \frac{1-p}{1-p_0} \Delta(H_p^\pi, p)$, and $\Delta(\tilde{G}, 1) := \Delta(H_p^\pi, 1) + \frac{p-p_0}{1-p_0} \Delta(H_p^\pi, p)$. By construction $\tilde{G} \succeq_{mps} H_p^\pi$ and $(\tilde{G}, p_0)$ is an equilibrium profile. The conclusion follows from noting that $H_{p_0}^{\pi_0} = \tilde{G}$, with $\pi_0 := p_0(1 - \tilde{G}(0)) < p(1 - H_p^\pi(0))$. $\square$

The above lemma identifies a relationship between the market price p and the equilibrium producer surplus π under which the designer can construct information policies that lead to strictly better consumer surplus.

Lemma 3. *For any efficient equilibrium profile (G, p) with outcome (κ, π) there exists a unique $p_\pi \geq p$ such that $t(p_\pi, \pi) = p_\pi$, $H_{p_\pi}^\pi \succeq_{mps} G$, and $(H_{p_\pi}^\pi, p_\pi)$ is an equilibrium profile.*

Moreover, for any equilibrium profile (G, p) the pair $(H_{p_\pi}^\pi, p_\pi)$ is also an outcome equilibrium profile with the same producer surplus and $H_{p_\pi}^\pi \succeq_{mps} G$.

Proof. By [Lemma 2](#) we get that (H_p^π, p) is an equilibrium profile such that $H_p^\pi \succeq_{mps} G$. If $p < t(p, \pi)$, then consider some $\varepsilon > 0$ such that $p + \varepsilon = t(p + \varepsilon, \pi)$. By [Theorem 1](#) we get that $t(p, \pi) > t(s, \pi)$ for all $s > p$ whenever it is well defined, thus an $\varepsilon > 0$ that satisfies the required equality exists. The uniqueness follows from the fact that [Equation 1](#) has a unique solution with $t(p, \pi) = p$. In particular, this

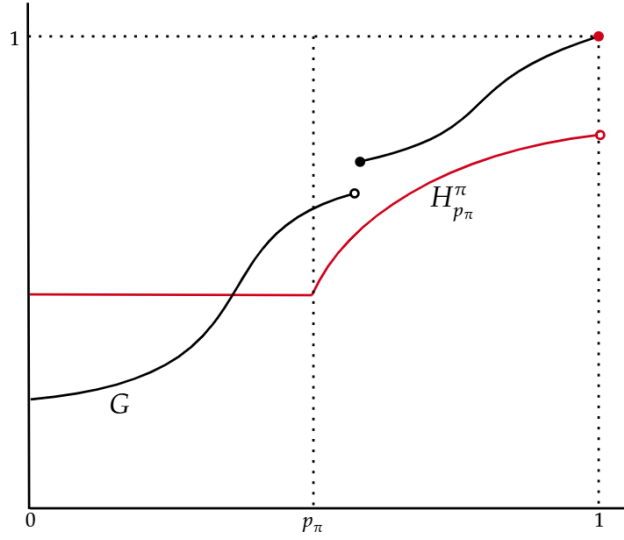


Figure 4: Single crossing between $H_{p_\pi}^\pi$ in black and some generic equilibrium demand G in red.

solution is $p_\pi = \exp\left(1 - \frac{EF}{\pi}\right)$. Finally, note that any price $s > p_\pi$ can not be supported by an efficient equilibrium profile with producer surplus π as Equation 1 can not be satisfied.

To prove the second statement, consider an equilibrium profile (G, p) , not necessarily efficient, with outcome (κ, π) . As π is maximum producer surplus under demand G we get that $1 - G(s) + \Delta(G, s) \leq \pi/s$ for all $s \in [0, 1]$. In particular $G(s) \geq 1 - \frac{\pi}{s}$ for all $s \in [0, 1]$. As (κ, π) is an equilibrium outcome, it must be that $\pi \leq EF$. This implies that $p_\pi = \exp\left(1 - \frac{EF}{\pi}\right) \in (0, 1]$. By construction $EH_{p_\pi}^\pi = EF = EG$, combining this with the definition of $H_{p_\pi}^\pi$ we get that G crosses $H_{p_\pi}^\pi$ once and from below. Thus $H_{p_\pi}^\pi \succeq_{mps} G \succeq_{mps} F$ completing the proof. $\square$

Let $\pi^* := \inf \{ \pi \mid \exists p \text{ such that } (H_p^\pi, p) \text{ is an equilibrium profile} \}$, and let $p^* := p_{\pi^*}$.

In particular by Lemma 3 and Equation 1 we get that $p^* = \exp\left(1 - \frac{EF}{\pi^*}\right)$. The producer surplus corresponding to the buyer's optimal outcome is given by π^* and is a solution to the following

$$\inf_{\pi \in [0, 1]} \pi \quad \text{subject to} \quad \int_0^x \left(H_{p_\pi}^\pi(s) - F(s) \right) ds \geq 0 \quad \text{for all } x \in [0, 1]$$

Where

$$p_\pi = \exp \left(1 - \frac{\mathbf{E}F}{\pi} \right)$$

Theorem 2. *If an equilibrium profile (G, p) is consumer optimal then it is outcome equivalent to $(H_{p^*}^{\pi^*}, p^*)$ and $H_{p^*}^{\pi^*} \succeq_{mps} G$.*

Proof. By Lemma 3 we get that there exists some $p_1 \geq p$ such that $(H_{p_1}^\pi, p)$ is an equilibrium profile and $H_{p_1}^\pi \succeq_{mps} G$. As G is consumer optimal, it is also efficient by Lemma 1. Thus, it must be that (G, p) generates producer surplus π^* . Finally by Theorem 1 and Lemma 3, it must be that $p^* = t(p^*, \pi^*)$ and $H_{p^*}^{\pi^*} \succeq_{mps} G$. $\square$

Example : Let $F \sim \text{Unif}[0, 1]$. By Theorem 2 we get that the optimal consumer surplus is given by

$$\inf_{\pi \in [0, 1]} \pi \quad \text{subject to} \quad \int_0^x \left(H_{p_\pi}^\pi(s) - s \right) ds \geq 0 \quad \text{for all } x \in [0, 1]$$

After solving the above problem numerically, we get the consumer surplus maximizing level of $\pi^* \simeq 0.204$.

References

- Elton, J. and Theodore P. Hill, “Fusions of a probability distribution.,” *The Annals of Probability*, 1992.
- Gentzkow, Matthew and Emir Kamenica, “A Rothschild-Stiglitz approach to Bayesian persuasion,” *American Economic Review*, 2016, 106 (5), 597–601.
- Roesler, Anne-Katrin and Balázs Szentes, “Buyer-optimal learning and monopoly pricing,” *American Economic Review*, 2017, 107 (7), 2072–2080.
- Shaked, Moshe and J. George Shanthikumar, “Stochastic orders.,” Springer, 2007.